

III. A. Conditional Generative Modeling

$$\mathcal{D}_{\text{paired}} = \{(x^{(i)}, y^{(i)})\}$$

$$p_{\cdot|y}(x | y, z)$$

$$p_{\theta}(x | z)$$

$$q_{\phi}(z | x)$$

 π_0
 $\hat{\pi}$
 $\pi_{\cdot|y}$

$$\hat{\pi}(\cdot), \nabla_x \hat{\pi}(\cdot)$$

$$\mathcal{D}_{\text{train}} = \{x^{(i)}\}$$

I. Unnormalised Sampling

II. Generative Modeling

III.B Pathwise Conditioning

